

Yossi Abutbul

Computer Science Student | Full-Stack Developer | RF Test Automation
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SUMMARY

CS student and RF integrator with 6+ years of lab experience. I write the software around RF hardware: Python instrument drivers, FastAPI backends and React dashboards that replace slow manual tests. My load-pull system turned a **3-day qualification into about 8 minutes**. Outside work I build web apps and learning tools with TypeScript, React and Next.js.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · C · C++

Frontend: React · Next.js · Tailwind · Three.js · Plotly · Chart.js

Backend: FastAPI · REST APIs · Firebase · Firestore

Test & RF: PyVISA · SCPI · Spectrum & Network Analyzers · Power Sensors · NB-IoT · CAT-M · LoRa · LTE · BLE

AI: OpenAI API · Whisper · LLM Agents · Claude Code · Codex · n8n

Tools: Git · Arduino

WORK EXPERIENCE

RF & Electronics Integrator - Arad Technologies

May 2020 - Present

- Built **Test Console**, a test automation platform for power amplifier tests, load pull and RF sweeps. It drives a DC power analyzer, network analyzer, power sensor, motor-driven trombone tuner and an Arduino-controlled RF switch on a 3D-printed rig. **Took a 3-day manual qualification down to about 8 minutes**.
- Designed a React + FastAPI platform that runs RF tests over Bluetooth with power and spectrum analyzers and generates the test reports automatically. **Cut reporting time by over 50%**.
- Built an interactive web tool for **visualizing antenna radiation patterns** (2D polar and 3D surface) and RF metrics, so new designs are reviewed in the browser instead of plotted by hand.
- Built a current logger that records transmit bursts, so battery life is **measured, not estimated**.
- Built a browser-based LoRa gateway log viewer that plots RSSI, SNR and frequency over time, keeping customer production logs on the local machine.
- Led HW/SW **bring-up** of NB-IoT, CAT-M and LoRa modules, including FEM integration and Python validation.

MILITARY SERVICE

Operational Project Leader & RF Technician - IDF Intelligence Corps, Unit 81

Apr 2017 - Dec 2019

- Ran multidisciplinary RF projects end to end under tight operational deadlines.
- Built Python tools to log and analyze spectrum analyzer data in the field, and integrated and troubleshot advanced RF systems.

PROJECTS

Haparlamentor | Next.js · TypeScript · Fuse.js · Whisper | In progress yossiabutbul.github.io/Haparlamentor

Type a line from the sitcom "Haparlament" and jump to the exact episode and timestamp. Built a custom AI agent that scrapes the episodes and transcribes them with OpenAI Whisper to create the search index.

OPlanner | React · TypeScript · Firebase oplaner-one.vercel.app

Student planner that imports the university calendar and keeps courses, deadlines and exams in one dashboard.

Pipeline CPU Simulator | React · TypeScript · Vite yossiabutbul.github.io/Pipeline_CPU

Educational tool I built to help students learn how a CPU pipeline works. Step through five stages cycle by cycle and watch hazards, forwarding and stalls happen.

C++ Hero | Educational game cpp-hero.vercel.app

Teaches modern C++ from zero as a game: 16 worlds, 802 challenges and a boss per world, with a focus on safe, defensive code.

EDUCATION

BSc Computer Science - The Open University of Israel · GPA 84

Oct 2022 - Oct 2027 (estimated)

Research paper (in progress): "Creating User Interfaces Using LLMs: From Specification to Code"